V8 Phase 10 Bell-Type Semantic Contextuality Pack

Date: 2026-04-23

Objective

Phase 10 locks the Bell-type semantic contextuality protocol on the compressed AI-feature state layer before hardware repeatability.

The protocol uses the existing V7 contextuality lane plus the locked V8 Phase 5 bridge features to define the semantic label buckets A, A', B, and B', then resolves family-local measurement vectors from the compressed two-qubit correlation matrix.

Protocol Boundary

- semantic / AI-side contextuality only
- compressed two-qubit feature states derived from locked public-safe summary features
- matched unentangled control uses the same single-qubit rotations with the entangling CX removed
- this is not a physical Bell nonlocality claim

Semantic Setting Buckets

- A: lattice / mirror-context setting
- A': matched alternate-context setting
- B: target-span bridge setting
- B': readout carry-through setting

Focused Cohorts

- full semantic overlap cohort: Mistral, Hermes, Gemma, GLM, Nemotron, Qwen, DeepSeek
- V8 state-only sidecar: SmolLM3

Aggregate Read

- mean entangled S_max: 2.018908
- mean matched-control S_max: 2.000000
- rows above classical bound: 8 / 8
- strongest model: DeepSeek at 2.040519
- weakest model: Gemma at 2.005572

First Read

Phase 10 locks the Bell-type semantic contextuality rung at the compressed state-geometry level. All compressed semantic feature states show S_max > 2 under resolved family-local measurement settings, while the matched unentangled controls remain exactly at the classical bound. This is a candidate semantic contextuality result on the AI-side feature-state layer, not a physical Bell nonlocality claim. The next step is hardware semantic repeatability with the locked setting map.

Per-Model Read

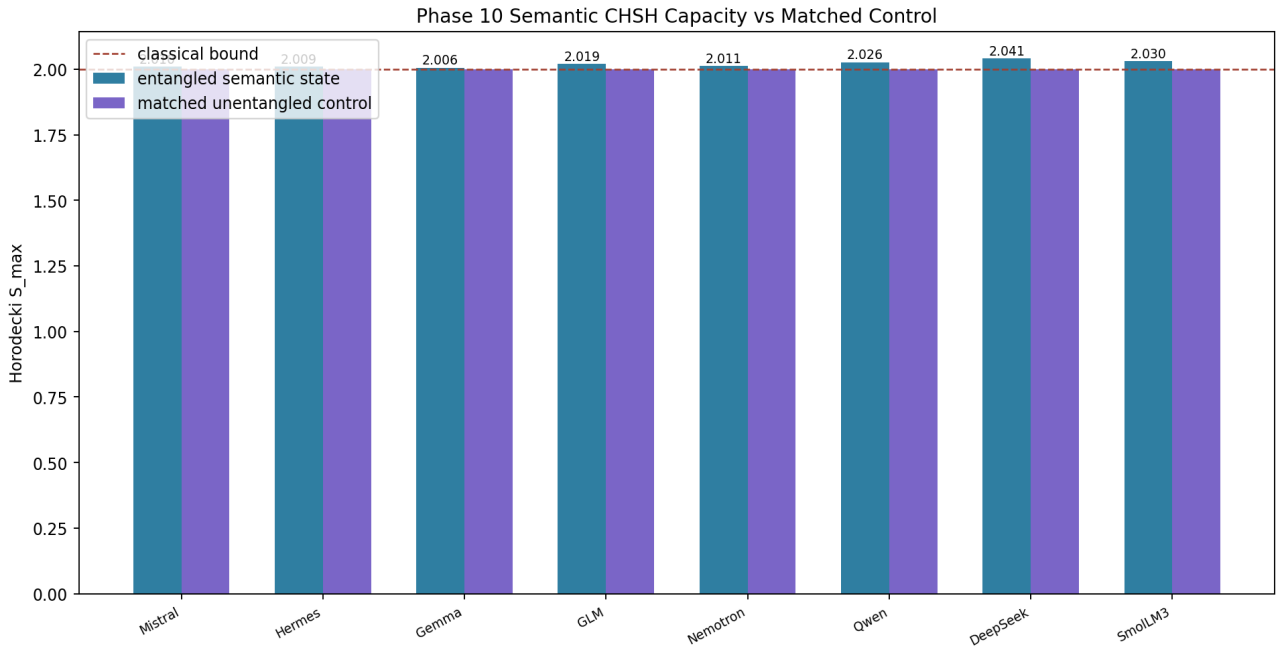
- Mistral (full_semantic_overlap): entangled S_max 2.009513, control 2.000000, entropy 0.024254
- Hermes (full_semantic_overlap): entangled S_max 2.009253, control 2.000000, entropy 0.023682
- Gemma (full_semantic_overlap): entangled S_max 2.005572, control 2.000000, entropy 0.015262
- GLM (full_semantic_overlap): entangled S_max 2.019500, control 2.000000, entropy 0.044825
- Nemotron (full_semantic_overlap): entangled S_max 2.011360, control 2.000000, entropy 0.028254
- Qwen (full_semantic_overlap): entangled S_max 2.025713, control 2.000000, entropy 0.056659
- DeepSeek (full_semantic_overlap): entangled S_max 2.040519, control 2.000000, entropy 0.083035
- SmolLM3 (v8_state_sidecar): entangled S_max 2.029830, control 2.000000, entropy 0.064218

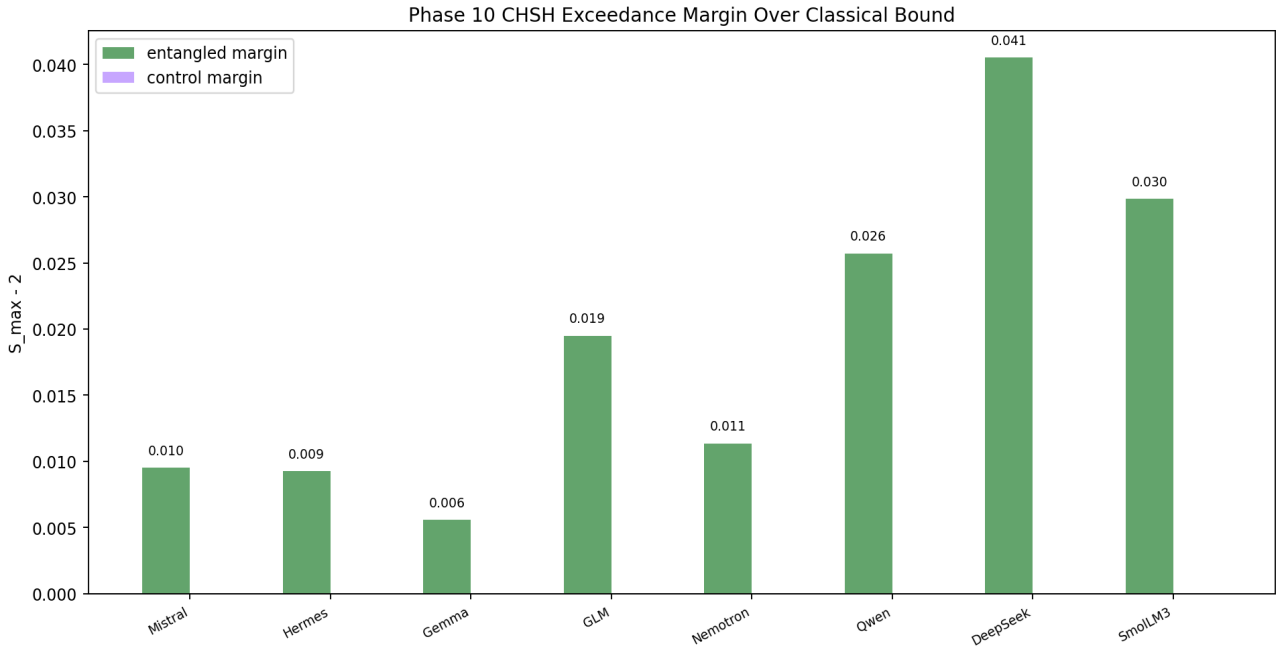
Charts

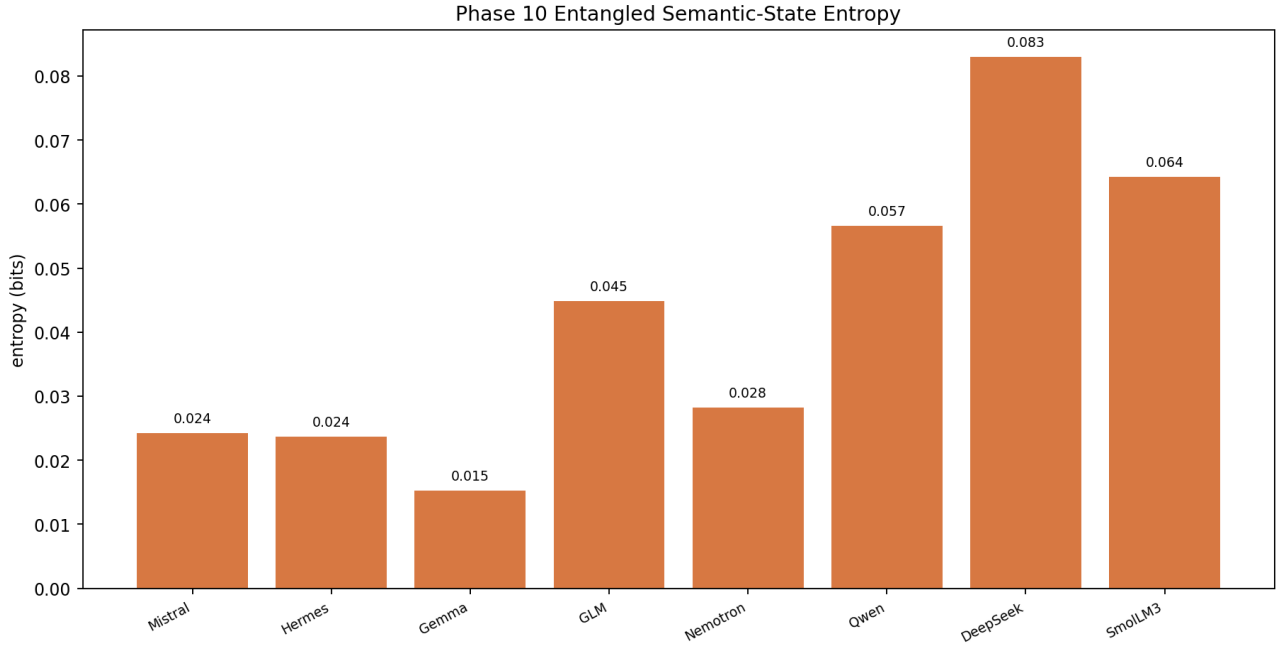
- [Semantic CHSH vs control](./charts/v8_phase10_semantic_chsh_vs_control_2026-04-23.png)
- [Exceedance margin](./charts/v8_phase10_semantic_exceedance_margin_2026-04-23.png)
- [Entanglement entropy](./charts/v8_phase10_semantic_entanglement_entropy_2026-04-23.png)

Next

The next rung is Phase 11: carry the locked Phase 10 semantic settings into real IBM hardware repeatability through both the Qiskit and direct PennyLane paths.




Phase 10 CHSH Exceedance Margin Over Classical Bound
entangled margin
control margin
S_max - 2
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.010
0.009
0.006
0.019
0.011
0.026
0.041
0.030
Mistral
Hermes
Gemma
GLM
Nemotron
Qwen
DeepSeek
SmolLM3


Phase 10 Entangled Semantic-State Entropy
entropy (bits)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.024
0.024
0.015
0.045
0.028
0.057
0.083
0.064
Mistral
Hermes
Gemma
GLM
Nemotron
Qwen
DeepSeek
SmolLM3